

# DevOps Lab

## Documentation

### **Program 4:**

Integrate Git with Docker for Source-Controlled Application Builds

The project involves integrating Git with Docker to enable source-controlled application builds. This ensures automated, version-controlled and reproducible Docker images directly from a Git repository.

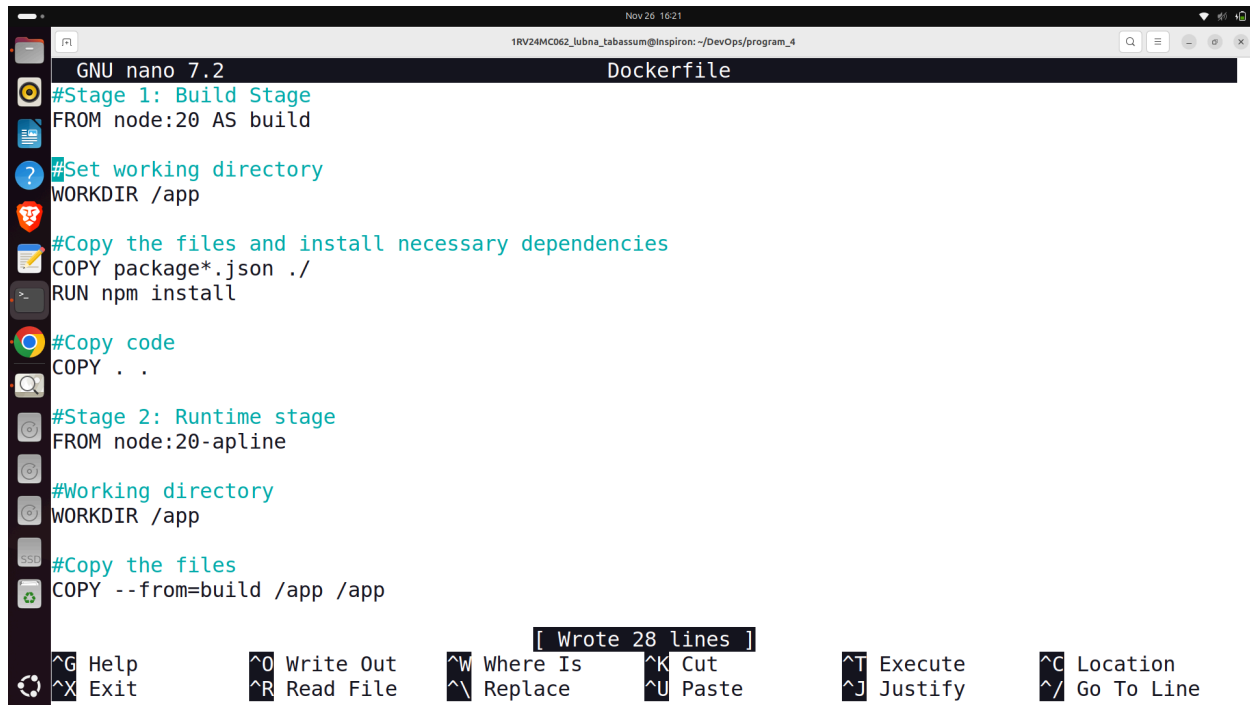
### **Project Structure:**

```
program_4/  
—>Dockerfile  
—>index.js  
—>package.json  
—>package-lock.json
```

### **Procedure:**

Create a folder (ex: program\_4)

**Step 1: Create a Dockerfile** without any extensions and add the following content  
nano Dockerfile



```
GNU nano 7.2 Dockerfile
#Stage 1: Build Stage
FROM node:20 AS build

#Set working directory
WORKDIR /app

#Copy the files and install necessary dependencies
COPY package*.json ./
RUN npm install

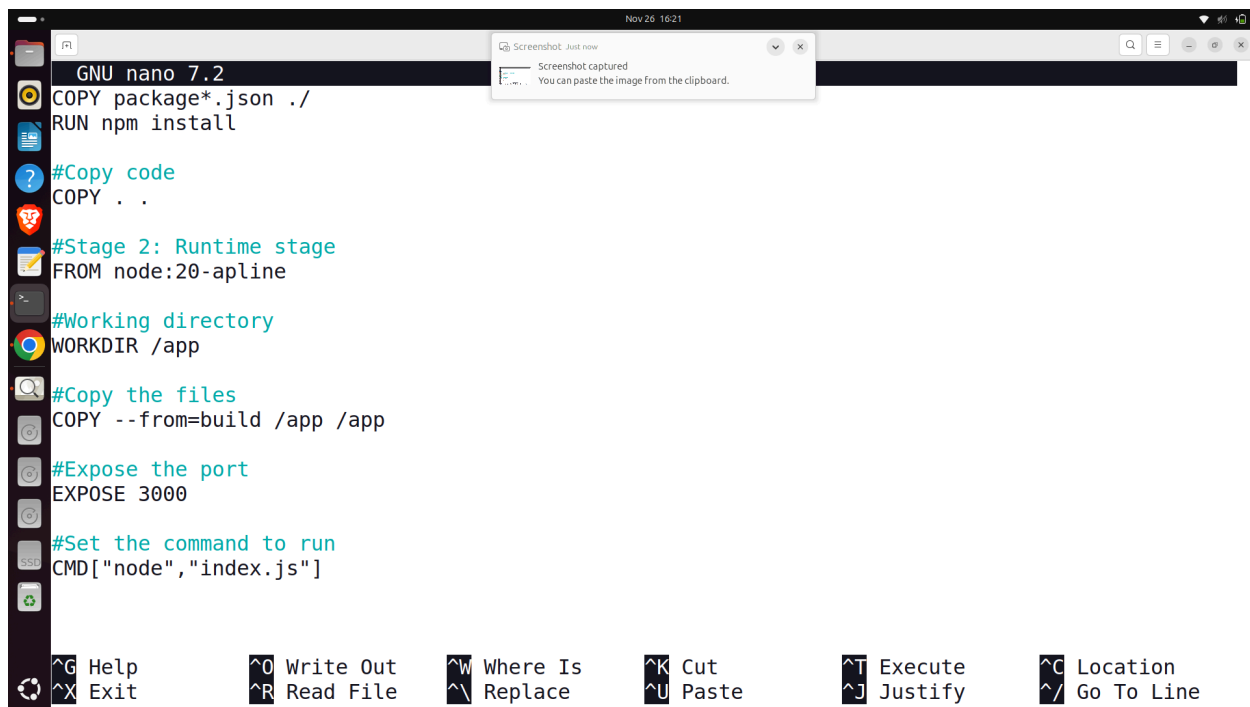
#Copy code
COPY . .

#Stage 2: Runtime stage
FROM node:20-apline

#Working directory
WORKDIR /app

#Copy the files
COPY --from=build /app /app

[ Wrote 28 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```



```
GNU nano 7.2
COPY package*.json ./
RUN npm install

#Copy code
COPY . .

#Stage 2: Runtime stage
FROM node:20-apline

#Working directory
WORKDIR /app

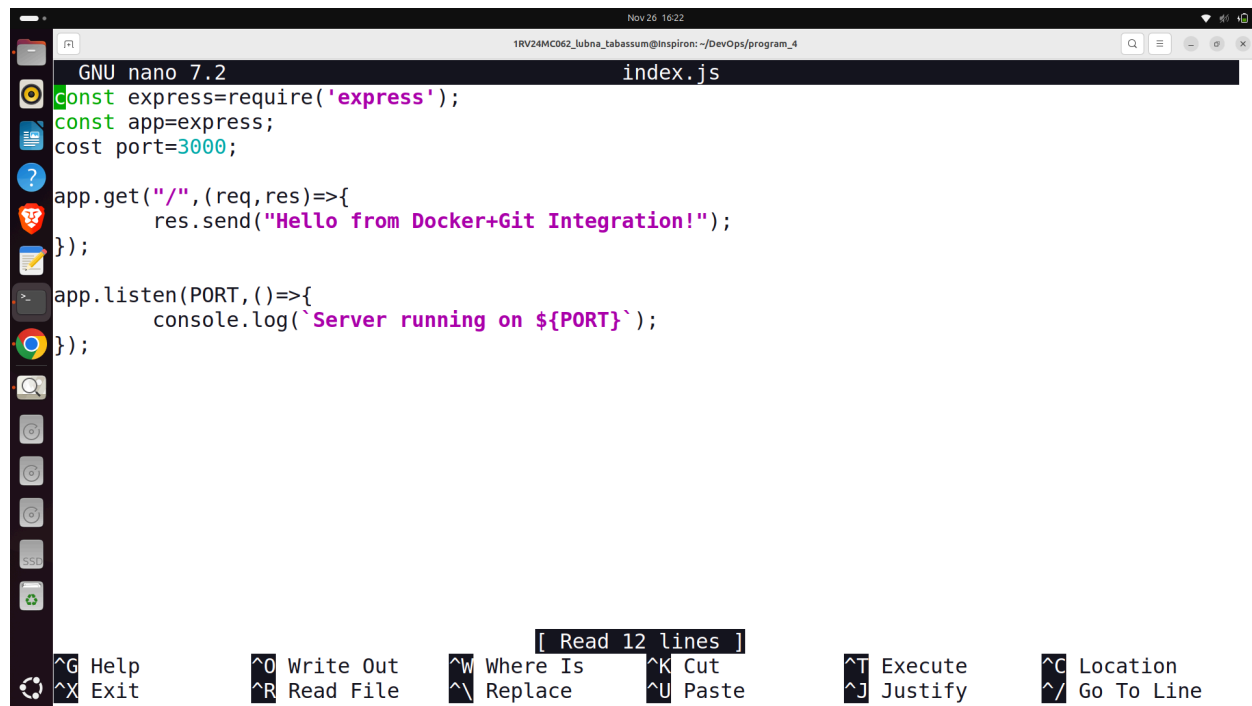
#Copy the files
COPY --from=build /app /app

#Expose the port
EXPOSE 3000

#Set the command to run
CMD ["node", "index.js"]

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

**Step 2:** Create an index.js file with following code  
nano index.js



```
GNU nano 7.2 index.js
const express=require('express');
const app=express;
const port=3000;

app.get("/",(req,res)=>{
    res.send("Hello from Docker+Git Integration!");
});

app.listen(PORT,()=>{
    console.log(`Server running on ${PORT}`);
});
```

[ Read 12 lines ]

|                |                     |                    |                 |                   |                      |
|----------------|---------------------|--------------------|-----------------|-------------------|----------------------|
| <b>^G</b> Help | <b>^O</b> Write Out | <b>^W</b> Where Is | <b>^K</b> Cut   | <b>^T</b> Execute | <b>^C</b> Location   |
| <b>^X</b> Exit | <b>^R</b> Read File | <b>^N</b> Replace  | <b>^U</b> Paste | <b>^J</b> Justify | <b>^_</b> Go To Line |

**Step 3:** create file package.json by executing the command  
npm init -y

```
Nov 19 16:13
1RV24MC062_lubna_tabassum@Inspiron: ~/DevOps/program_4
1RV24MC062_lubna_tabassum@Inspiron:~$ cd DevOps
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps$ mkdir program_4
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps$ cd program_4
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ nano Dockerfile
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ nano index.js
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ npm init
This utility will walk you through creating a package.json file.
It only covers the most common items, and tries to guess sensible defaults.

See `npm help init` for definitive documentation on these fields
and exactly what they do.

Use `npm install <pkg>` afterwards to install a package and
save it as a dependency in the package.json file.

Press ^C at any time to quit.
package name: (program_4) npm warn init canceled
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ npm init -y
Wrote to /home/1RV24MC062_lubna_tabassum/DevOps/program_4/package.json:

{
  "name": "program_4",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {}
}
```

#### Step 4: Initialize Git Repository locally.

```
git config --global user.name ""
git config --global user.email ""
git init
git add .
git commit -m "Initial commit"
```

```
Nov 19 16:13
1RV24MC062_lubna_tabassum@Inspiron: ~/DevOps/program_4

1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ git config --global user.name "LubnaTabassum-ux"
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ git config --global user.email "lubnata610@gmail.com"
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1RV24MC062_lubna_tabassum/DevOps/program_4/.git/
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ git add .
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ git commit -m "Initial commit"
[master (root-commit) 7935d4e] Initial commit
631 files changed, 72118 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
```

## Step 5: Link to remote repository

```
Nov 19 16:14
1RV24MC062_lubna_tabassum@Inspiron: ~/DevOps/program_4

1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ git remote -v
origin https://github.com/lubnatabassum-ux/hello.git (fetch)
origin https://github.com/lubnatabassum-ux/hello.git (push)
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ git push -u main
fatal: 'main' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ git push -u origin main
Username for 'https://github.com': lubnatabassum-ux
Password for 'https://lubnatabassum-ux@github.com':
Enumerating objects: 716, done.
Counting objects: 100% (716/716), done.
Delta compression using up to 8 threads
Compressing objects: 100% (661/661), done.
Writing objects: 100% (716/716), 723.16 KiB | 8.51 MiB/s, done.
Total 716 (delta 140), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (140/140), done.
remote: This repository moved. Please use the new location:
remote: https://github.com/LubnaTabassum-ux/hello.git
To https://github.com/lubnatabassum-ux/hello.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$
```

## Step 6: Push code to GitHub

`git push -u origin main`

and the repository exists.

```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ git push -u origin main
```

```
Username for 'https://github.com': lubnatabassum-ux
```

```
Password for 'https://lubnatabassum-ux@github.com':
```

```
Enumerating objects: 716, done.
```

```
Counting objects: 100% (716/716), done.
```

```
Delta compression using up to 8 threads
```

```
Compressing objects: 100% (661/661), done.
```

```
Writing objects: 100% (716/716), 723.16 KiB | 8.51 MiB/s, done.
```

```
Total 716 (delta 140), reused 0 (delta 0), pack-reused 0
```

```
remote: Resolving deltas: 100% (140/140), done.
```

```
remote: This repository moved. Please use the new location:
```

```
remote: https://github.com/LubnaTabassum-ux/hello.git
```

```
To https://github.com/lubnatabassum-ux/hello.git
```

```
* [new branch]      main -> main
```

```
branch 'main' set up to track 'origin/main'.
```

```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_4$ █
```

## Output

